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Course: STA1020A

Task: Quiz 2

A company uses three call center (A, B and C) to handle customer support, contributing 40%, 35% and 25% of the total calls, respectively. The probabilities that a call is unresolved are:

call center: A = 4%

B = 3%

C = 5%

(i) Probability that a randomly selected call is unresolved

Soln

Let call which is unresolved be X

$$\therefore P(X) = (P(A) \cdot P(X|A)) + (P(B) \cdot P(X|B)) + (P(C) \cdot P(X|C))$$

$$= (0.4 \times 0.04) + (0.35 \times 0.03) + (0.25 \times 0.05)$$

$$= 0.016 + 0.0105 + 0.0125$$

$$= 0.039$$

$\therefore$ The probability that the call is unresolved is 0.039 which is 3.9%

(ii) Probability that the call came from call center C

Soln

$$P(C|X) = \frac{P(C) \times P(X|C)}{P(X)}$$

$$\frac{0.25 \times 0.05}{0.039}$$

$$0.039$$

$$\frac{0.0125}{0.039} \approx 0.3205$$

$\therefore$ The probability that it came from C is

0.3205 which is 32.05%

2) In a health survey, it was found that:

65% of people exercise regularly (E)
50% follow a healthy diet (D)
40% do both

(i) What is the probability that the person either exercises or follows diet (or both)?

Soln
$$P(E \cup D) = P(E) + P(D) - P(E \cap D)$$

$$= 0.65 + 0.50 - 0.40$$

$$= 1.15 - 0.40$$

$$= 0.75$$

which is 75%

(ii) The probability that the person follows healthy diet but does not exercise

Soln

$$P(D \cap E') = P(D) - P(E \cap D)$$

$$= 0.50 - 0.40$$

$$= 0.1$$

which is 10%

) The probability that the person neither E nor D

Soln

$$P(E' \cap D') = 1 - P(E \cup D)$$

$$1 - 0.75$$

$$= 0.25$$

$$P(E' \cap D') = 1 - P(E \cup D)$$

$$1 - 0.75$$

$$= 0.25$$

which is 25%

3). A small bakery produces batches of cookies, where the number of defective cookies y in a randomly selected batch follows the probability mass function:

$$f(y) = \frac{1}{15}(y+1), y = 0, 1, 2, 3, 4.$$

(i) verify that the given function is a valid probability mass function

Soln

$$f(y) = \frac{1}{15}(y+1)$$

$$= \frac{y+1}{15}$$

$$\sum f(y) = \frac{y+1}{15} = 1$$

Substitute y in the function to find the summation = 1

$$\frac{0+1}{15} + \frac{1+1}{15} + \frac{2+1}{15} + \frac{3+1}{15} + \frac{4+1}{15}$$

$$\frac{1}{15} + \frac{2}{15} + \frac{3}{15} + \frac{4}{15} + \frac{5}{15}$$

$$= \frac{15}{15}$$

$$= 1$$

Therefore probability mass function is valid

(ii) The probability that at least 3 defective cookies

Soln

$$P(Y \geq 3) = f(3) + f(4)$$

$$\frac{3+1}{15} + \frac{4+1}{15}$$

$$\frac{4}{15} + \frac{5}{15}$$

$$\frac{9}{15}$$

$$= 0.6 \text{ which is } 60\%$$

(iii) Suppose the bakery discards any batch with more than 2 defective cookies. What is the probability of discarding a randomly selected batch?

Soln
 $P(Y > 2) = P(Y = 3) + P(Y = 4)$

$$\frac{3+1}{15} + \frac{4+1}{15}$$

$$\frac{4}{15} + \frac{5}{15}$$

$$\frac{9}{15}$$

which is 0.6
60%

i) calculate $E(Y)$

Soln

$$E(Y) = \sum y \times f(y)$$

$$= \left(0 \times \frac{0+1}{15}\right) + \left(1 \times \frac{1+1}{15}\right) + \left(2 \times \frac{2+1}{15}\right) + \left(3 \times \frac{3+1}{15}\right) + \left(4 \times \frac{4+1}{15}\right)$$

$$= \left(0 \times \frac{1}{15}\right) + \left(1 \times \frac{2}{15}\right) + \left(2 \times \frac{3}{15}\right) + \left(3 \times \frac{4}{15}\right) + \left(4 \times \frac{5}{15}\right)$$

$$= 0 + \frac{2}{15} + \frac{6}{15} + \frac{12}{15} + \frac{20}{15}$$

$$= \frac{40}{15} = \frac{8}{3}$$

$$= \frac{8}{3}$$

$$= 2.67$$

(v) Calculate variance $\text{var}(Y)$

$$\text{var}(Y) = E(Y^2) - (E(Y))^2$$

$$E(Y^2) = \sum y^2 \times f(y)$$

$$= (0^2 \times \frac{1}{15}) + (1^2 \times \frac{2}{15}) + (2^2 \times \frac{3}{15}) + (3^2 \times \frac{4}{15}) + (4^2 \times \frac{5}{15})$$

$$= 0 + \frac{2}{15} + \frac{12}{15} + \frac{36}{15} + \frac{80}{15}$$

$$= \frac{130}{15}$$

$$= 8.67$$

$$(E(Y))^2 = (2.67)^2$$

$$= 7.1289$$

$$\text{var}(Y) = 8.67 - 7.1289$$

$$= 1.5411$$

$$= 1.54 \text{ (2 d.p.)}$$